

Feature 1:

Chosen Values:

Material: PLA

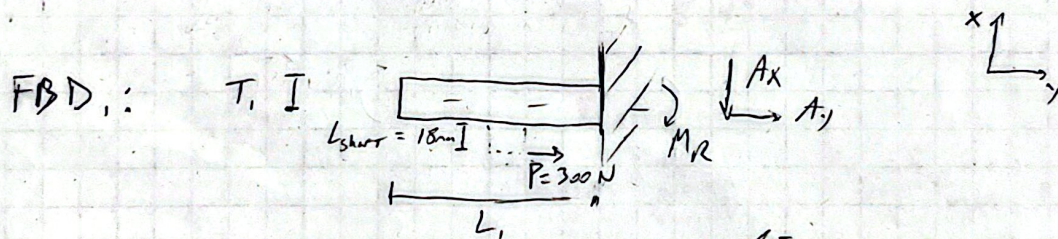
- $\sigma_{yield} = 45.2 \text{ MPa}$ (MatWeb)
- $E = 2350 \text{ MPa}$ (MatWeb)
- $L_1 = 50 \text{ mm}$
- $W_1 = 30 \text{ mm}$

KNOWN VALUES:

- Applied Load $P = 300 \text{ N}$
- Deflection $\Delta_{max} = 0.30 \text{ mm}$
- Safety Factor (FOS) = 3
- $L_{shaft} = 18 \text{ mm}$

Unknown Values:

- Thickness (T_1)
- Moment (M_1)



$$\sum F_y = 0 = P - A_x$$
$$\boxed{P = A_x}$$

$$\sum M = 0 = (P L_1) - M_R$$
$$\boxed{M_R = P L_{shaft}}$$

$$M_R = M_1 = P L_{shaft}$$

$$M_1 = (300 \text{ N})(18 \text{ mm}) = \boxed{5400 \text{ N}\cdot\text{mm}}$$

Allowable Axial Stress: (σ_{allow})

$$FOS = \frac{\sigma_{yield}}{\sigma_{allow}}$$

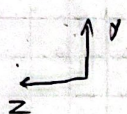
$$\rightarrow \sigma_{allow} = \frac{\sigma_{yield}}{FOS}$$

$$\sigma_{allow} = \frac{(45.2 \text{ MPa})}{(3)}$$

$$\boxed{\sigma_{allow} = 15.07 \text{ MPa}}$$

Feature 1 (cont.):

Cross-Sectional Area with Moment of Inertia (I):



$$T_1 \quad w_1 \rightarrow I_1 = \frac{w_1 (T_1)^3}{12}$$

Stress Limit: Flexure Formula

$$\frac{\sigma_{yield}}{FOS} = \sigma_{allow} = \frac{M_1 (C_1)}{I_1} = \frac{PL_{shear} \left(\frac{T_1}{2} \right)}{\left(\frac{w_1 (T_1)^3}{12} \right)} = \frac{6M_1}{w_1 \cdot T_1^2}$$

$$T_{1,r} = \sqrt{\frac{6M_1}{\left(\frac{\sigma_{yield}}{FOS} \right) \cdot w_1}} \rightarrow T_{1,r} = \sqrt{\frac{6(P L_{shear})}{\left(\frac{\sigma_{yield}}{FOS} \right) \cdot w_1}} \quad (eq. 1)$$

$$T_{1,r} = \sqrt{\frac{6(5400 \text{ N}\cdot\text{mm})}{(15.07 \text{ MPa})(30 \text{ mm})}}$$

$$T_{1,r} = 8.47 \text{ mm}$$

Beam Deflection Equation:

$$\Delta L = \frac{M_1 L_1^2}{2EI_1} = \frac{(P L_{shear}) L_1^2}{2E \left(\frac{w_1 (T_1)^3}{12} \right)}$$

$$T_{1,\Delta L} = \sqrt[3]{\frac{6P L_{shear} (L_1)^2}{E \cdot w_1 \cdot \Delta L}} \quad (eq. 2)$$

$$T_{1,\Delta L} = \sqrt[3]{\frac{6(300 \text{ N})(16 \text{ mm})(50 \text{ mm})^2}{(2350 \text{ MPa})(30 \text{ mm})(0.30 \text{ mm})}}$$

$$T_{1,\Delta L} = 15.65 \text{ mm}$$

$$15.65 \text{ mm} > 8.47 \text{ mm}$$

$$\boxed{\text{Governed by } 15.65 \text{ mm} = T_1}$$

Feature 2:

Chosen Values:

• Maintain width $\rightarrow W_1 = W_2 = 30\text{ mm}$

Knowns: • $\Delta L_{\text{max}} = 0.30\text{ mm}$

• $FOS = 3$

• $P = 300\text{ N}$

• $\sigma_{\text{allow}} = 15.07\text{ MPa}$

• Clearance Hole for Bolt ($d_B = 3.4\text{ mm}$)

• $L_{\text{sheet}} = 19\text{ mm}$

Unknowns:

• Length ($L_{2\text{TOTAL}}$)

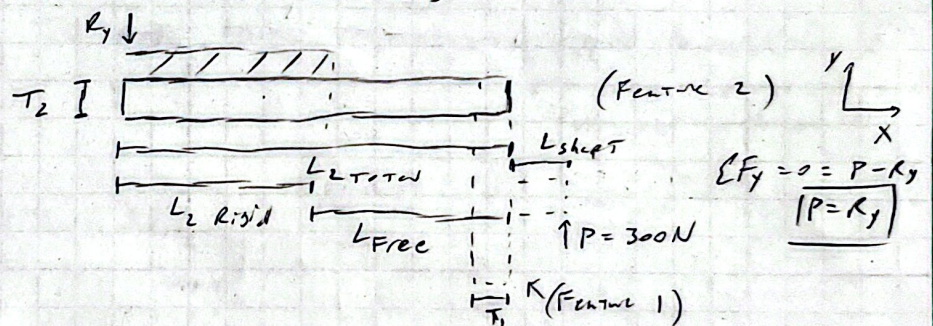
• Length (L_{Rigid})

• Length (L_{Free})

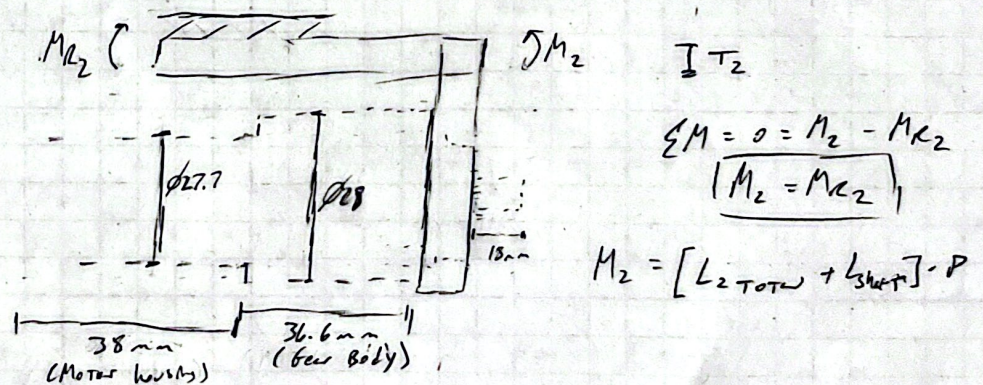
• Thickness (T_2)

FBD₂:

(1)



(2)



Determining $L_{2\text{TOTAL}}$: $L_{2\text{TOTAL}} = (\text{Gear Body}) + (\text{Motor Housing}) + T_1$

$$L_{2\text{TOTAL}} = (36.6\text{ mm}) + (38\text{ mm}) + (15.65\text{ mm})$$

$$L_{2\text{TOTAL}} = 90.25\text{ mm} \approx 91\text{ mm}$$

$$M_2 = [L_{2\text{TOTAL}} + L_{\text{sheet}}] \cdot P$$

$$M_2 = (91\text{ mm} + 19\text{ mm}) \cdot (300\text{ N}) = 32700\text{ N}\cdot\text{mm}$$

Feature 2 (cont): Fixing L_{2sid}

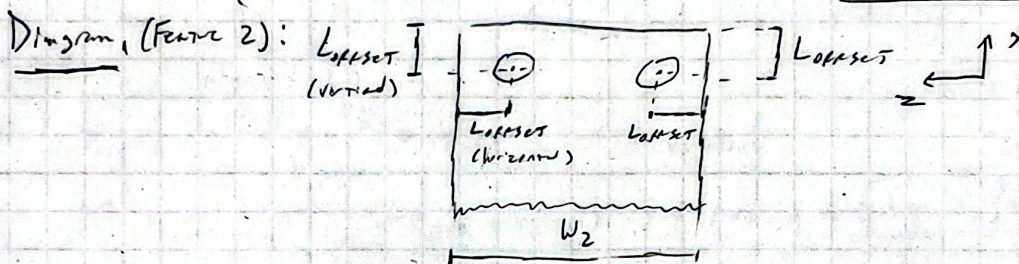
Since Clearance For Bolt holes are given: $\varnothing$
 $(d_B = 3.4\text{mm})$

Able To use Structural Engineering Standard:

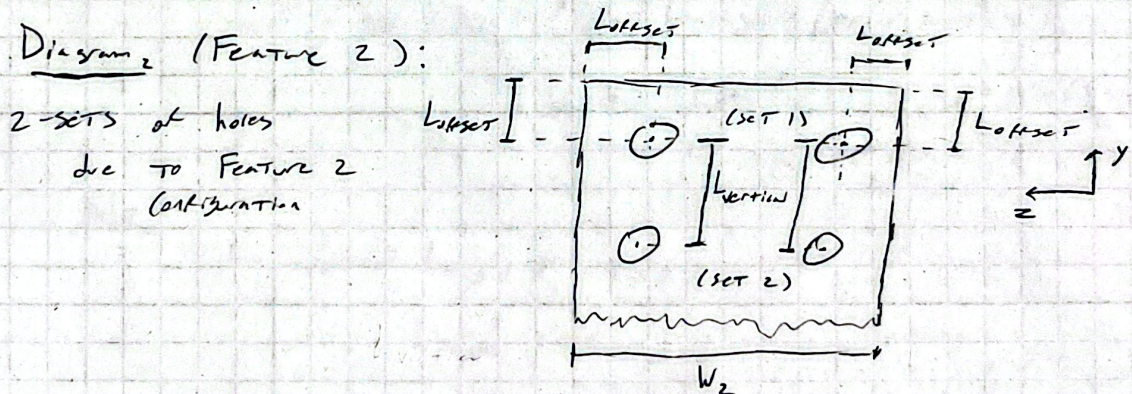
(Distance From center of hole to closest edge = L_{offset})

$$L_{\text{offset}} = 2 \cdot d_B \quad (\text{From respective edge})$$

$$L_{\text{offset}} = 2 \cdot (3.4\text{mm}) = 6.8\text{mm} \approx 7\text{mm} = L_{\text{offset}}$$



To make Holes more rigid: Apply L_{offset}
 To Both distances
 (Vertical & Horizontal)



To Find L_{vertical}: Due TO Structural Engineering Standard!

$$\left(\text{Center-to-Center distance From one SET TO another} \right) = \text{"Minimum Bolt Pitch Rule"} = L_{\text{vertical}}$$

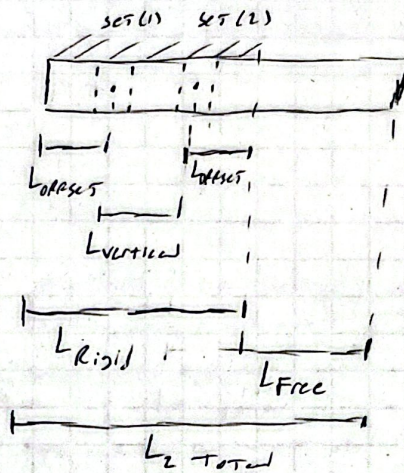
Minimum Bolt Pitch Rule: $L_{\text{vertical}} = 4 \cdot d_B$

$$L_{\text{vertical}} = 4(3.4\text{mm}) = 13.6\text{mm}$$

$L_{\text{vertical}} \approx 20\text{mm}$
 Bigger distance between
 Sets = more leverage TO
 resist motion

Feature 2 (L₂): Solving For L_{Rigid}

Diagram:



$$L_{RIGID} = L_{OFFSET} (2) + L_{VERTICAL}$$

$$L_{RIGID} = (7mm)(2) + (20mm) = \boxed{34mm}$$

Solving For L_{Free}:

$$L_{2\ TOTAL} = (\text{gear housing}) + (\text{motor body}) + T_1$$

$$L_{2\ TOTAL} = (36.6mm) + (38mm) + (15.65mm)$$

$$L_{2\ TOTAL} \approx 91mm$$

Also:

$$L_{2\ TOTAL} = L_{RIGID} + L_{FREE}$$

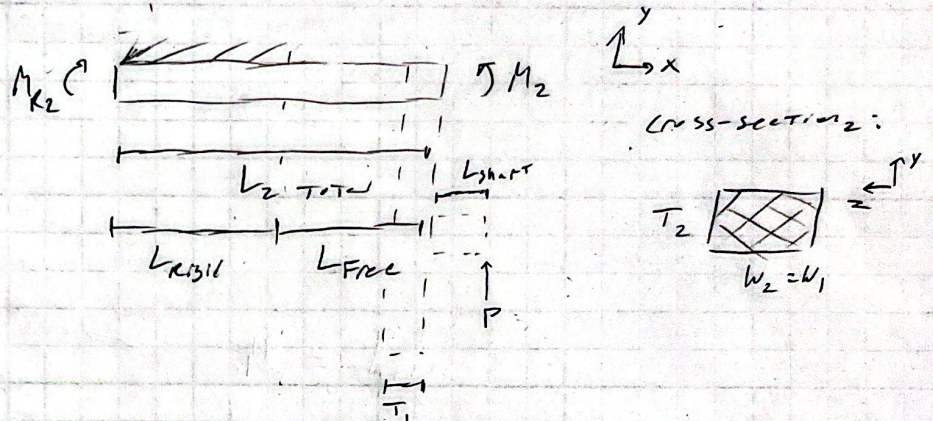
$$L_{FREE} = L_{2\ TOTAL} - L_{RIGID}$$

$$L_{FREE} = (91mm) - (34mm) = \boxed{57mm}$$

Feature 2 (cont.):

Now That We Solved For L_{Free}
We use This Bell's Length To calculate moment (T_2)

FBD₂:



T_2 Based on Flexure Formula:

$$\sigma_{allow} = \frac{M_2 (c_2)}{I_2}$$

$$\text{Moment of Inertia } I_2 = \frac{w_2 (T_2)^3}{12}$$

$$\frac{\sigma_{yield}}{FOS} = \sigma_{allow} = \frac{M_2 \left(\frac{T_2}{2} \right)}{\left(\frac{w_2 (T_2)^3}{12} \right)}$$

$$T_2 = \sqrt{\frac{6 (L_{2Total} + L_{2Short}) (P)}{\left(\frac{\sigma_{yield}}{FOS} \right) w_2}} \quad (\text{eq 1})$$

$$T_{2s} = \sqrt{\frac{6 (32700 \text{ N}\cdot\text{mm})}{(15.07 \text{ MPa}) (30 \text{ mm})}}$$

$$T_{2Flex} = T_2 = 20.832 \text{ mm}$$

T_2 Based on Deflection:

$$\Delta L = \frac{M_2 (L_{Free})^2}{2 E I_2} = \frac{[(L_{2Total} + L_{2Short}) (P)] (L_{Free})^2}{2 E \left(\frac{w_2 (T_2)^3}{12} \right)}$$

$$(\text{eq 2}) \quad T_{2dL} = \sqrt[3]{\frac{6 M_2 (L_{Free})^2}{E \cdot w_2 \cdot \Delta L}} = \sqrt[3]{\frac{6 [(L_{2Total} + L_{2Short}) (P)] (L_{Free})^2}{E \cdot w_2 \cdot \Delta L}}$$

$$T_{2dL} = \sqrt[3]{\frac{6 (32700 \text{ N}\cdot\text{mm}) (57 \text{ mm})^2}{(2350 \text{ MPa}) (30 \text{ mm}) (0.30 \text{ mm})}}$$

$$T_{2dL} = 31.12 \text{ mm}$$

$T_{2Flex} < T_{2dL} \rightarrow$ Governing Thickness

VALUES

Chosen/Given:

Material: PLA

$$\sigma_{yield} = 45.2 \text{ MPa} \quad (\text{MatWeb})$$

$$E = 2350 \text{ MPa} \quad (\text{MatWeb})$$

$$\Delta L_{max} = 0.30 \text{ mm}$$

$$P = 300 \text{ N}$$

$$FOS = 3$$

Feature 1	Symbolic	Value
τ_{allow}	$= \frac{\sigma_{yield}}{FOS}$	15.07 MPa
$T_{1\sigma}$	$= \sqrt{\frac{6(P \cdot L_{short})}{(\frac{\sigma_{yield}}{FOS}) \cdot W_1}}$	8.47 mm
$T_{1\Delta L}$	$= \sqrt[3]{\frac{6(P \cdot L_{short})(L_1)^2}{E \cdot W_1 \cdot \Delta L}}$	15.65 mm
M_1	$= L_1 (P)$	5400 N·mm

$L_1 = 50 \text{ mm}$
 $W_1 = W_2 = 3 \text{ mm}$

Feature 2	Symbolic	Value
L_{2TOTAL}	$= (\text{Gear Body}) + (\text{Motor Housing}) + T_1$ $= L_{2Rigid} + L_{2Free}$	$\sim 9 \text{ mm}$
L_{offset}	$= 2 \cdot (d_g)$	$\sim 7 \text{ mm}$
$L_{vertical}$	$= 4 \cdot (d_g)$	$\sim 20 \text{ mm}$
L_{Rigid}	$= 2(L_{offset}) + L_{vertical}$	34 mm
L_{Free}	$= L_{2TOTAL} - L_{Rigid}$	57 mm
M_2	$= P(L_{2TOTAL})$	32700 N·mm
$T_{2Flex} = T_{2\sigma}$	$= \sqrt{\frac{6(L_{2TOTAL} \cdot L_{short})(P)}{(\frac{\sigma_{yield}}{FOS}) \cdot W_2}}$	20.732 mm

VALUES CONT.

Feature 2 (cont.)

	Symbolic	VALUE
$T_{2\Delta L}$	$\sqrt[3]{\frac{6[(L_{2TOT} + L_{2H2})^2(P)] \cdot \Delta L_{free}}{E \cdot W_2 \cdot \Delta L}}$	31.12 mm

ISO - view

$$W_2 = W_1 = 30 \text{ mm}$$

$$31.12 = T_2$$

$$T_{\text{an}} = L_{\text{an}} = 15.65 \text{ mm}$$

$$\phi_{\text{ds}} = 3.4 \text{ mm}$$

Latent

$$L_{\text{vertical}} = 29 \text{ mm}$$

$$L_{\text{TOTAL}} = 71 \text{ mm}$$

$$L_1 = 50 \text{ mm}$$

$$T_1 = 15.65 \text{ mm}$$

$$W_1 = W_2 = 30 \text{ mm}$$